		$I_1 = 0.413I_0 \times 0.587 = 0.24I_0$
14	**D**	$I = kA^2$ $\frac{I_1}{I_2} = (\frac{A_1}{A_2})^2$ $\frac{4I_2}{I_2} = (\frac{A_1}{A})^2$ $A_1 = 2A$ If amplitude of S_2 is A, then amplitude of S_1 is 2A. Resultant amplitude at P, $A_P = A_1 + A_2 = 2A + A = 3A$ Resultant amplitude at Q, $A_P = A_1 - A_2 = 2A - A = A$ Ratio of Intensity $\frac{I_P}{I_Q} = \frac{A_p{}^2}{A_Q{}^2} = \frac{(3A)^2}{(A)^2} = 9.0$
15	**C**	$a\sin\theta = n\lambda$